Week 6 Worksheet — Blood Pressure Regression

Health Mini-Project (3σ Preprocessing & Machine Learning Blocks)

Name: _____ **Date:** _____

Story Setup — Cleaning and Modeling Real Health Data
By now we know that data are never perfect. This week we focus on **detecting and removing outliers** before modeling. We also connect all the small code blocks (#%%) into one full regression pipeline for predicting **Systolic (SBP)** and **Diastolic (DBP)** blood pressure.

Goals of This Week

- Compute basic statistics (mean, standard deviation, min, max) for each numeric column.
- Apply the **3σ rule**: keep only data within $\mu \pm 3\sigma$ for each column.
- Print and discuss values that are physically unrealistic (e.g., SBP=999).
- Understand the full ML pipeline through ten #%% blocks.
- Relate the five core ideas of learning to the correct code blocks.

A. Code Blocks Overview (10 total)

1. **Setup & Reproducibility** — paths, random seed, configuration.
   Why: ensures experiments are repeatable.
2. **Load Data & Schema Map** — read CSV, rename columns, pick features/targets.
   Why: a consistent schema avoids downstream errors.
3. **Missingness & Basic Stats Report** — show total rows/cols, NA counts, min/max.
   Why: visualize possible data-entry mistakes.
4. **3σ Outlier Cutoff (Report → Filter)** — compute μ, σ, print how many rows fall outside each bound.
   Why: remove obvious typos/outliers for clean learning.
5. **Numericization & Feature Engineering** — convert buckets, compute BMI/WHtR, encode categories.
   Why: models require numerical input.
6. **Train/Validation Split** — divide data 75/25 with fixed seed.
   Why: keep validation data unseen for fair testing.

7. **Scaling (Standardization)** — fit scaler on train, apply to validation only.
   Why: avoid information leakage.

8. **Models (Linear vs. MLP)** — define structure, hidden nodes, activation.
   Why: compare linear vs. nonlinear representations.

9. **Training + Staged Search** — gradient descent with step LR; coarse → refined hyper-parameter search.
   Why: efficient exploration; note local minima risk.

10. **Evaluation & Reporting** — compute RMSE/MAE/R^2, residual checks, data kept ratio.
    Why: interpret model and data quality together.

B. Preprocessing Report (fill in after code run)

Initial rows (total)	_____
Columns (total)	_____
Rows containing any NA (to drop)	_____
Rows after NA removal	_____
Rows inside $\pm 3\sigma$ bounds (all numeric cols)	_____
Rows removed by 3σ cutoff	_____
Kept ratio within 3σ (%)	_____

Tip: For each numeric column, note min/max before filtering and highlight any impossible values (e.g., negative waist size).

C. The "Core Five" Concepts

1. Normalize / Numericize Data
2. Linear Combination
3. Hidden Nodes
4. Nonlinear Activation
5. Staged Search + Gradient Descent (Local Minima Risk)

D. Task 1 — Match the Core 5 to the Code Blocks

Write the corresponding #%% block numbers (multiple answers allowed):

A. Normalize / Numericize Data	_____
B. Linear Combination	_____
C. Hidden Nodes	_____
D. Nonlinear Activation	_____
E. Staged Search + Gradient Descent	_____

Hint: Scaling is Block 7; model structure is Block 8; GD and search are Block 9.

E. Task 2 — Try Changing One Hyper-Parameter
When you receive the full code, each student will modify one hyper-parameter and record its effect.

Hyper-param	Options / Range	Your Choice	Metric Change	Explanation
Learning rate (LR)	1e-2, 3e-3, 1e-3			
Hidden size (MLP)	16, 32, 64			
Activation	relu, tanh			
Epochs	100, 200, 400			
Batch size	32, 64, 128			
Regularization (linear)	none, ridge(0.1/1.0)			
Outlier σ-k	2.5, 3.0, 3.5			
Scaler	standard, minmax			

Note: Adjusting the σ cutoff changes how many rows remain, thus altering both data quality and quantity.

F. Teacher Notes (for Discussion)

- **3σ Rule:** Simple and visual; discuss why μ and σ themselves can be distorted by outliers.
- **Alternative Methods:** Median/MAD, IQR, winsorization — preview for future lessons.
- **Local Minima:** Nonlinear models may get stuck; multi-seed and staged search help.
- **Data Leakage:** Fit scalers only on training data to keep evaluation honest.
- **Reporting:** Always show before/after counts and per-column min/max.

Turn-in Checklist

Preprocessing summary table (rows/columns, NA, 3σ kept ratio).

Core 5 $\leftrightarrow$ block matching answers.

One hyper-parameter experiment and short explanation.

Reflection: Why is cleaning before modeling essential?